

# **DSP Controller for Embedding Multiple Control Of EV Battery Charging**



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# Contents:

- 1.Introduction
- 2.System Overview
- 3.MPPT-Based Solar Charging System
- 4.PSFB-Based Grid Charging System
- 5.DSP Controller Implementation
- 6.Integrated System Operation
- 7.Experimental Results
- 8.Conclusion & Future Scope

# Introduction:

- This project develops an integrated EV battery charging system using a DSP controller.
- It combines solar PV charging via MPPT-controlled boost converter
- Integrates grid charging via phase-shift full-bridge converter
- Implements real-time control algorithms on TMS320F28379D DSP
- Creates a unified charging platform supporting multiple energy sources

# Project Motivation

Key Challenges in EV Charging:

Solar PV efficiency varies with environmental conditions

Grid charging requires high power factor and efficiency

Multiple controllers increase system cost and complexity

Need for adaptive charging algorithms

## Our Solution

Single DSP controlling multiple power stages

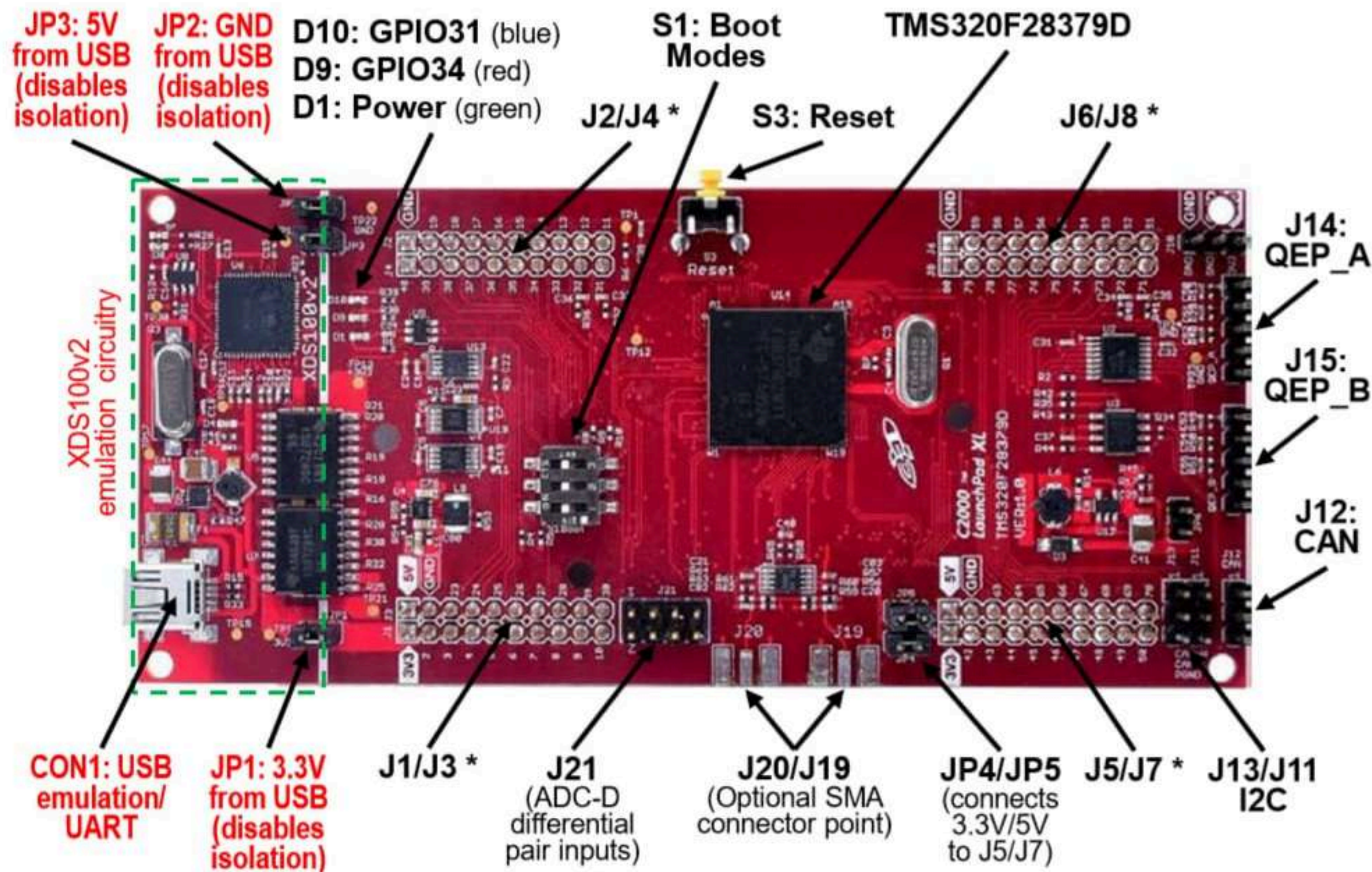
Real-time algorithm execution

Seamless source switching

High efficiency across operating modes



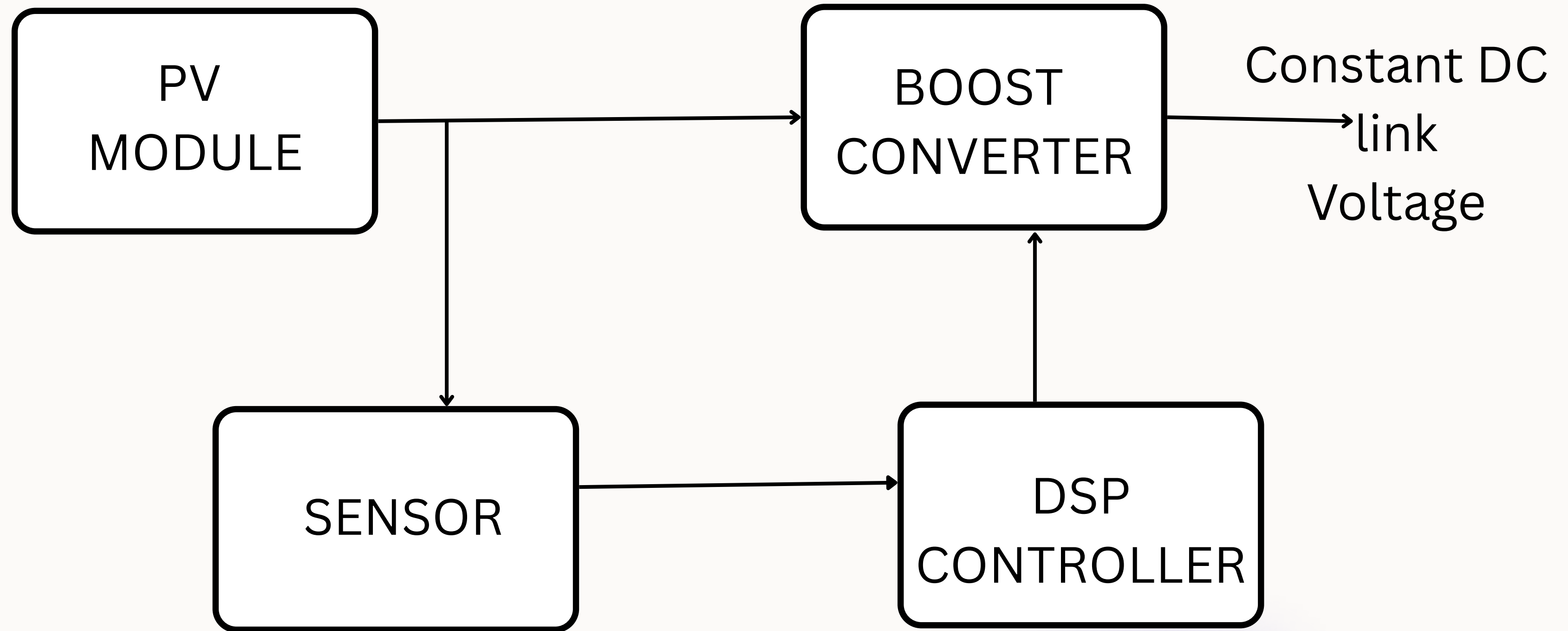
# F28379D LaunchPad



\* = BoosterPack plug-in module connector      Note: F28379D – 337 pin package



# MPPT Control model:



# Real-Time Maximum Power Point Tracking:

## How It Works:

The DSP continuously monitors solar panel output through voltage and current sensors, calculating instantaneous power. Using the Perturb & Observe (P&O) algorithm, it adjusts the boost converter's duty cycle to track the Maximum Power Point (MPPT).

# Sensor Integration:

- Voltage Sensor: Measures PV panel voltage in real-time
- Current Sensor: Monitors PV panel current continuously
- ADC Conversion: DSP converts analog readings to digital values
- Power Calculation: DSP computes  $P = V \times I$  every sampling cycle



## **Algorithm Execution:**

The DSP compares current power with previous measurements, deciding whether to increase or decrease duty cycle to move toward higher power output. This continuous adjustment compensates for changing sunlight conditions.

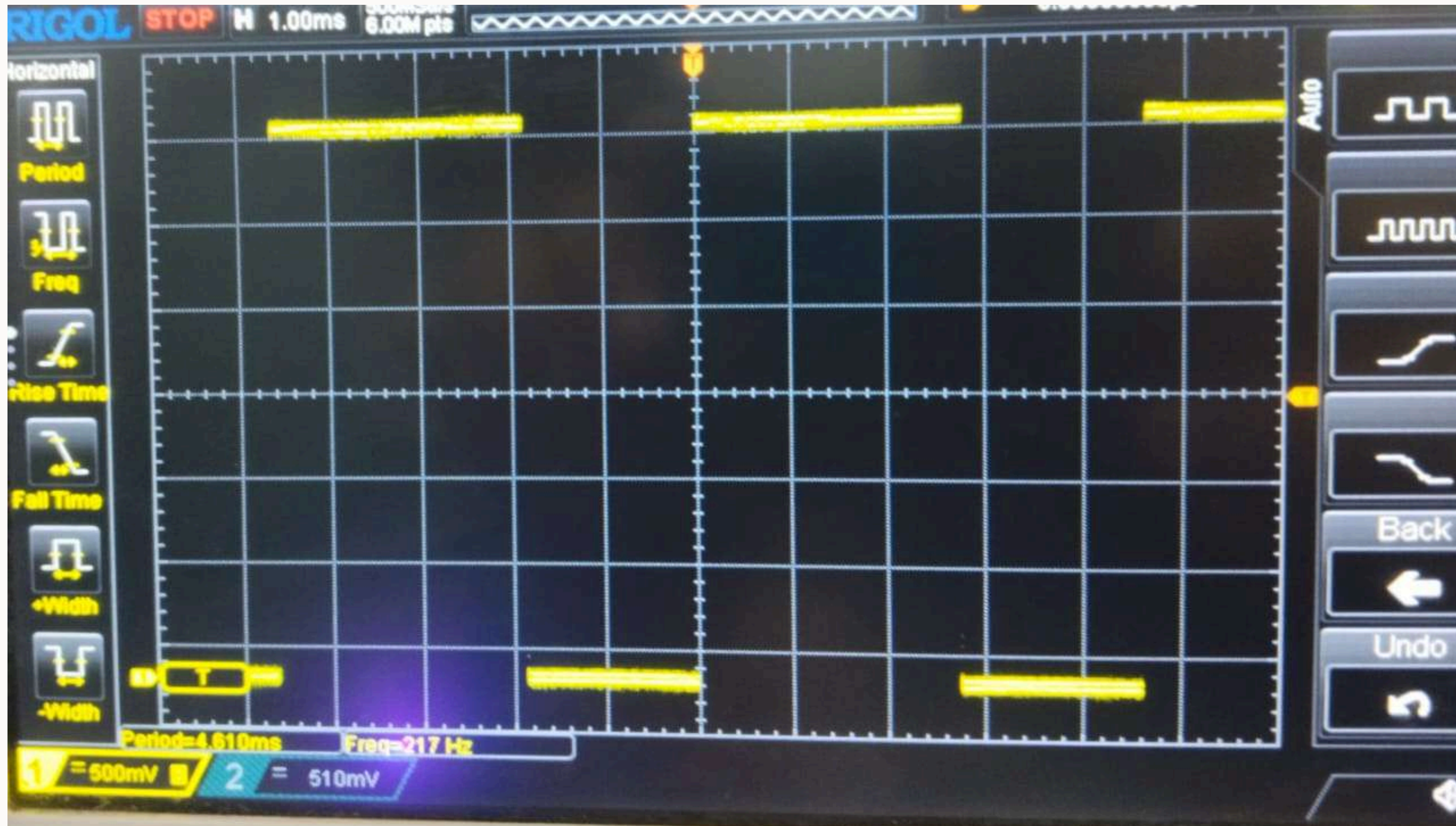
# P&O MPPT Algorithm Implementation:

Algorithm Flow:

1. Measure  $V_{pv}(k)$  and  $I_{pv}(k)$
2. Calculate  $P_{pv}(k) = V_{pv}(k) \times I_{pv}(k)$
3. Compute  $\Delta P = P(k) - P(k-1)$   
Compute  $\Delta V = V(k) - V(k-1)$
4. Decision Logic:
  - If  $\Delta P > 0$  and  $\Delta V > 0 \rightarrow$  Increase Duty
  - If  $\Delta P > 0$  and  $\Delta V < 0 \rightarrow$  Decrease Duty
  - If  $\Delta P < 0$  and  $\Delta V > 0 \rightarrow$  Decrease Duty
  - If  $\Delta P < 0$  and  $\Delta V < 0 \rightarrow$  Increase Duty
5. Update PWM Duty Cycle
6. Repeat every 100ms

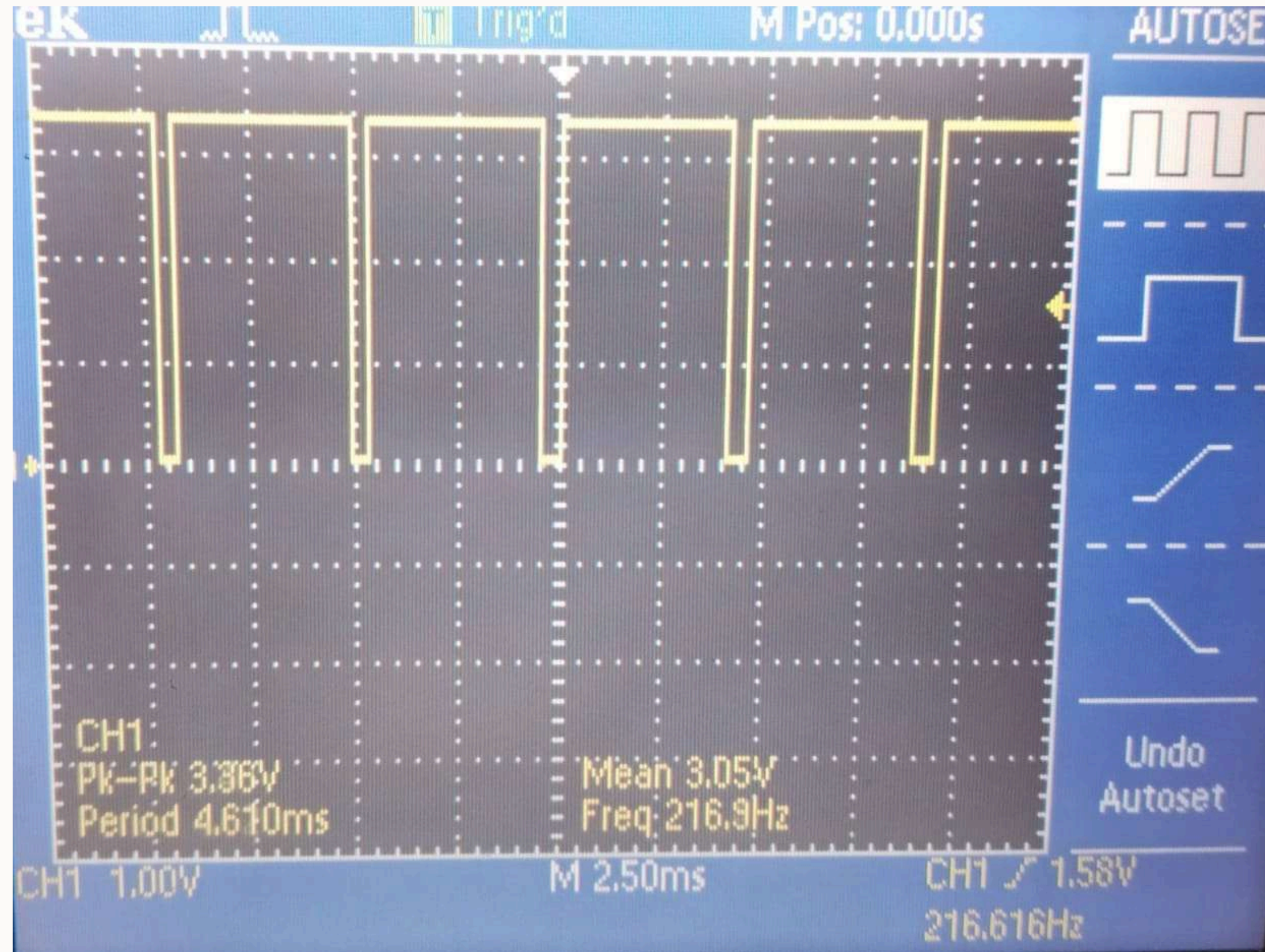


# Waveform for Boost Converter control:





# MPPT Control Duty Ratio Waveform:



## **ACHIEVED OUTCOME:**

Successfully established a constant DC link from variable solar input through a DSP-controlled Boost Converter with real-time sensor feedback, enabling stable power delivery to subsequent charging stages.



# PSFB Implementation with IGBTs & DSP Control:

## **How DSP Enables This:**

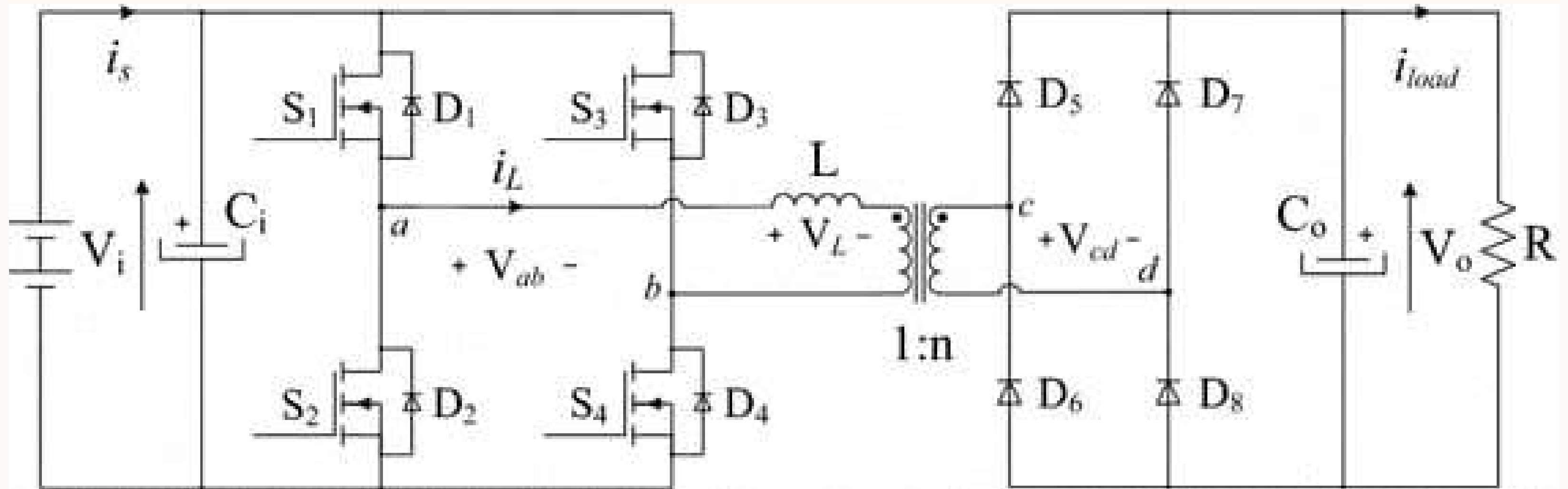
The DSP creates precisely timed phase differences between the two bridge legs of the PSFB converter. By shifting the switching timing between left and right legs, power transfer is controlled without changing the duty cycle.

## **Phase-Shift Concept:**

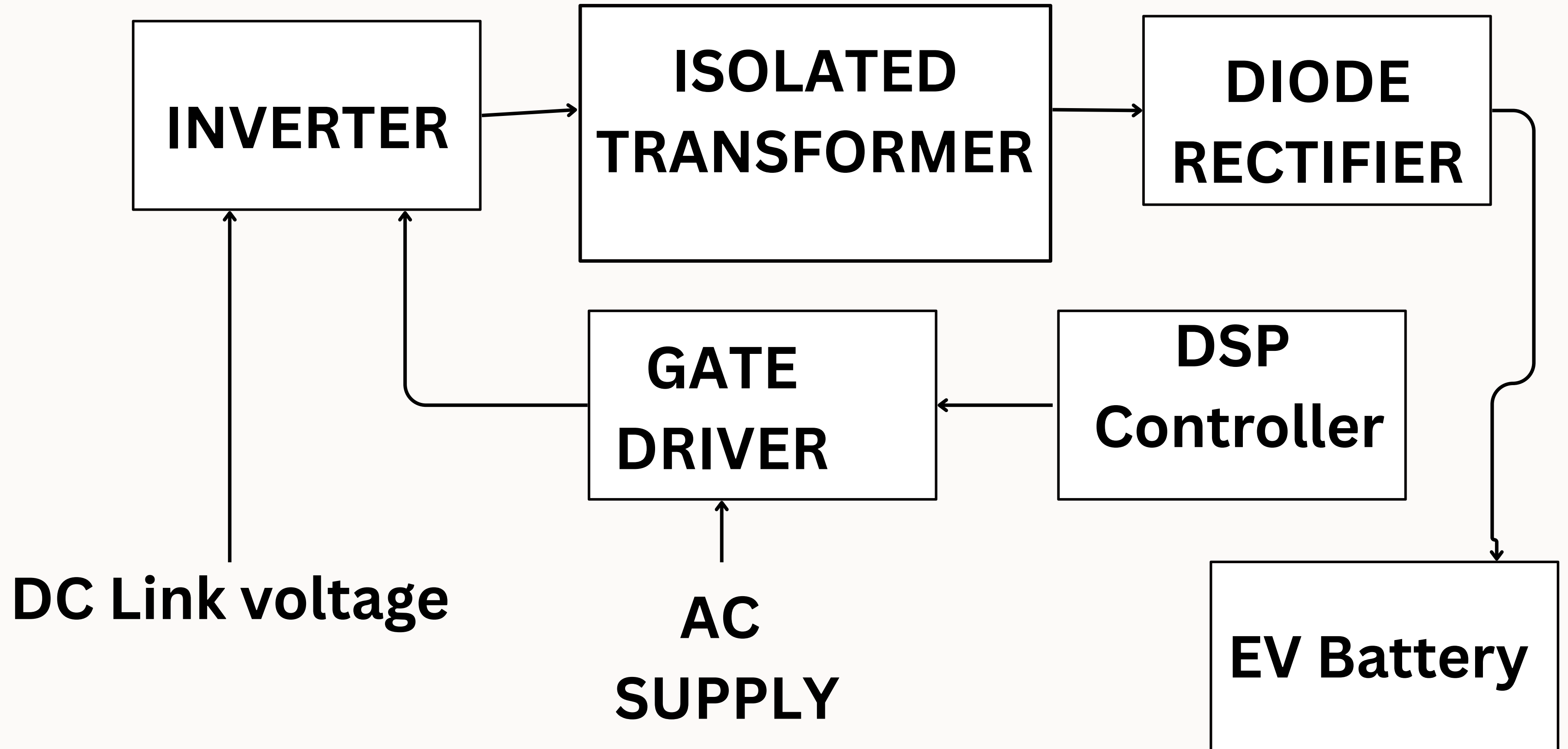
The DSP's PWM modules generate four synchronized signals with adjustable phase relationships. These signals drive the IGBTs through gate drivers, creating alternating diagonal conduction paths through the transformer.



## PSFB Circuit Diagram:



# PSFB Converter Model:



# **PSFB Phase-Shift Control with DSP:**

## **Bridge Leg Operation & Timing Control**

Leg 1 & Leg 2 Configuration:

The PSFB converter consists of two bridge legs controlled independently by the DSP:

Leg 1: Comprises IGBTs Q1 (high-side) and Q2 (low-side)

Leg 2: Comprises IGBTs Q3 (high-side) and Q4 (low-side)



# Dead-Band Implementation:

The DSP inserts programmable dead-bands between complementary switches within the same leg:

Between Q1 and Q2 in Leg 1

Between Q3 and Q4 in Leg 2

Purpose: Prevents shoot-through current by ensuring one switch turns OFF completely before the other turns ON.

## Phase-Shift Control:

The DSP applies a controlled phase shift between Leg 1 and Leg 2:

- Both legs operate at 50% duty cycle
- Leg 2 switching is delayed relative to Leg 1 by a phase angle ( $\varphi$ )
- This phase difference determines power transfer to the transformer
- Varying  $\varphi$  from  $0^\circ$  to  $180^\circ$  controls output voltage/current

## Working Sequence:

Diagonal Conduction Phase: Q1 and Q4 conduct simultaneously → current flows through transformer

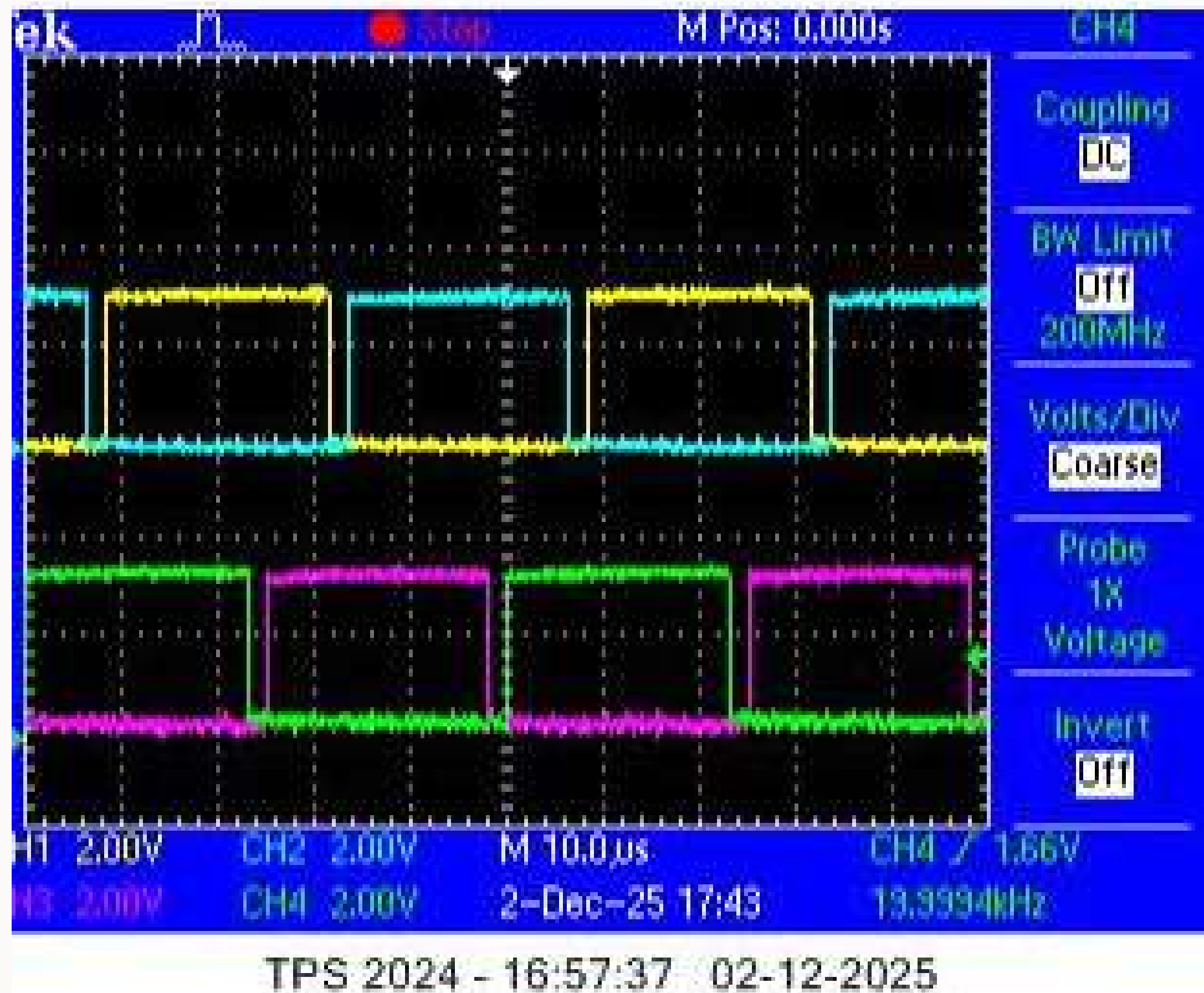
- Freewheeling Phase: All switches OFF, energy circulates in resonant components
- Complementary Diagonal Phase: Q2 and Q3 conduct → reverse current through transformer
- Freewheeling Phase: Energy reset completes switching cycle

## **DSP's Role:**

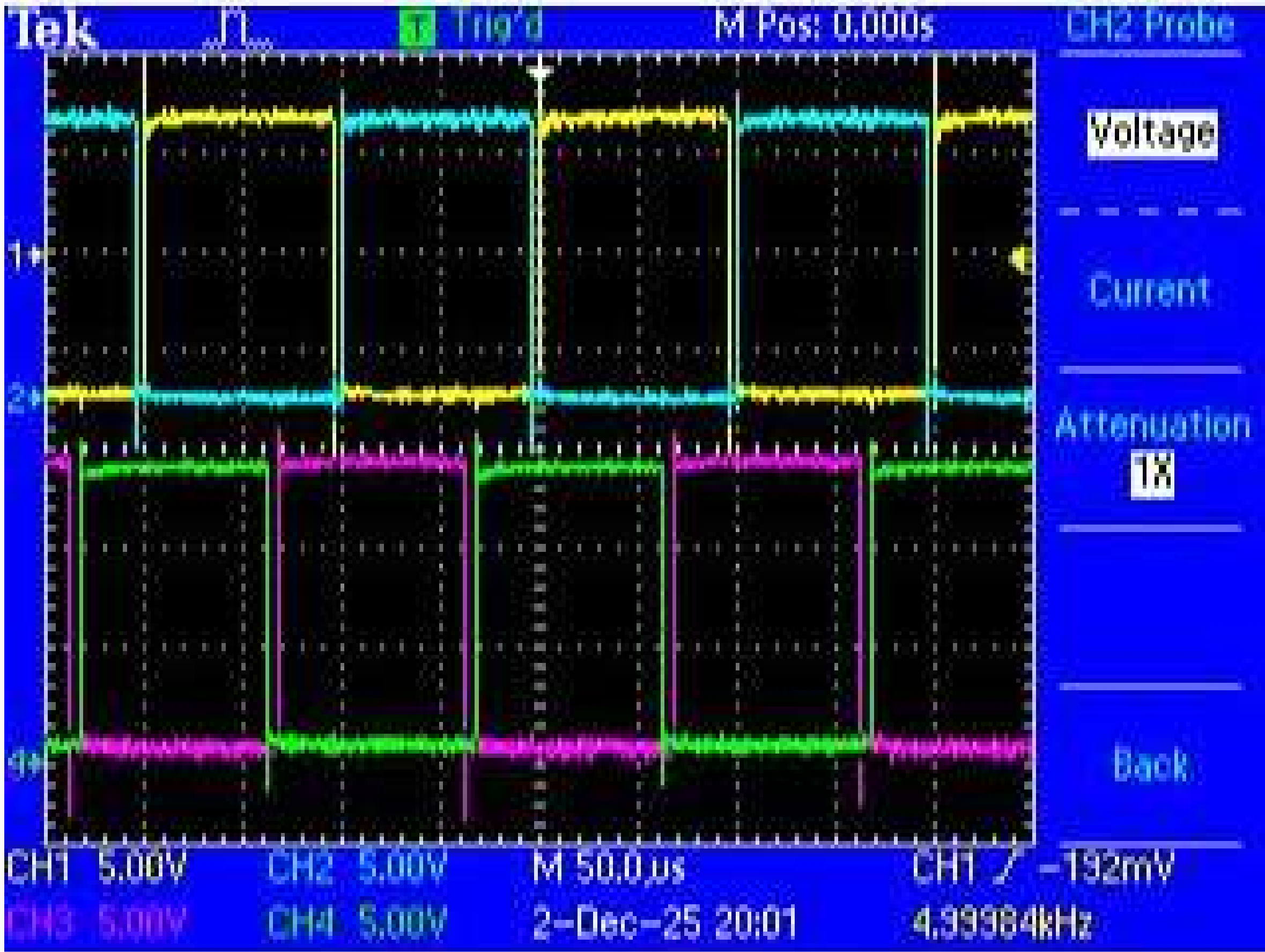
- Precisely controls phase shift ( $\varphi$ ) between legs
- Manages dead-band timing for safe switching
- Adjusts phase in real-time based on load demand
- Ensures soft-switching conditions are maintained



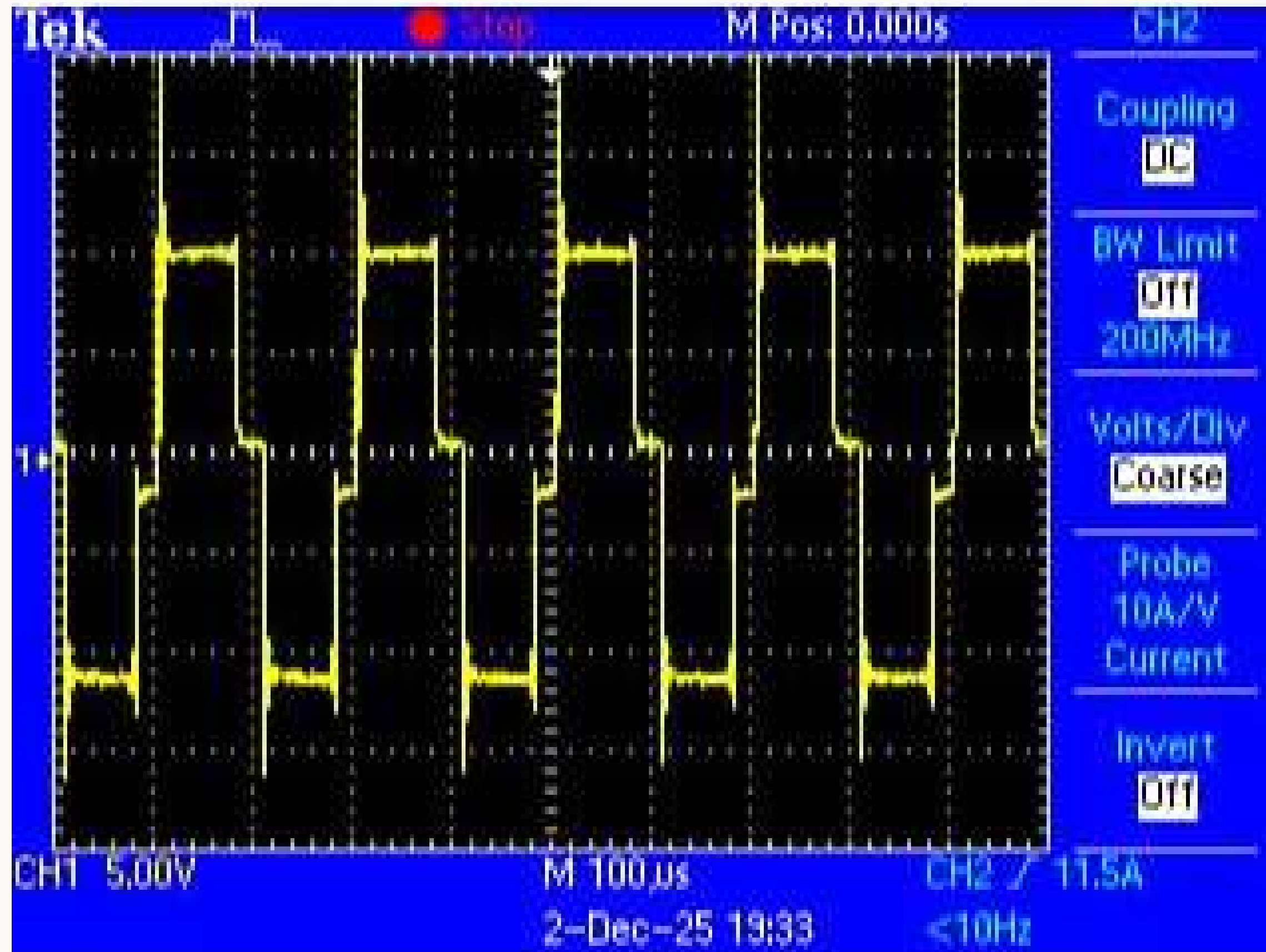
# DSP output waveform for Gate Driver:



# Gate Driver Output waveform:

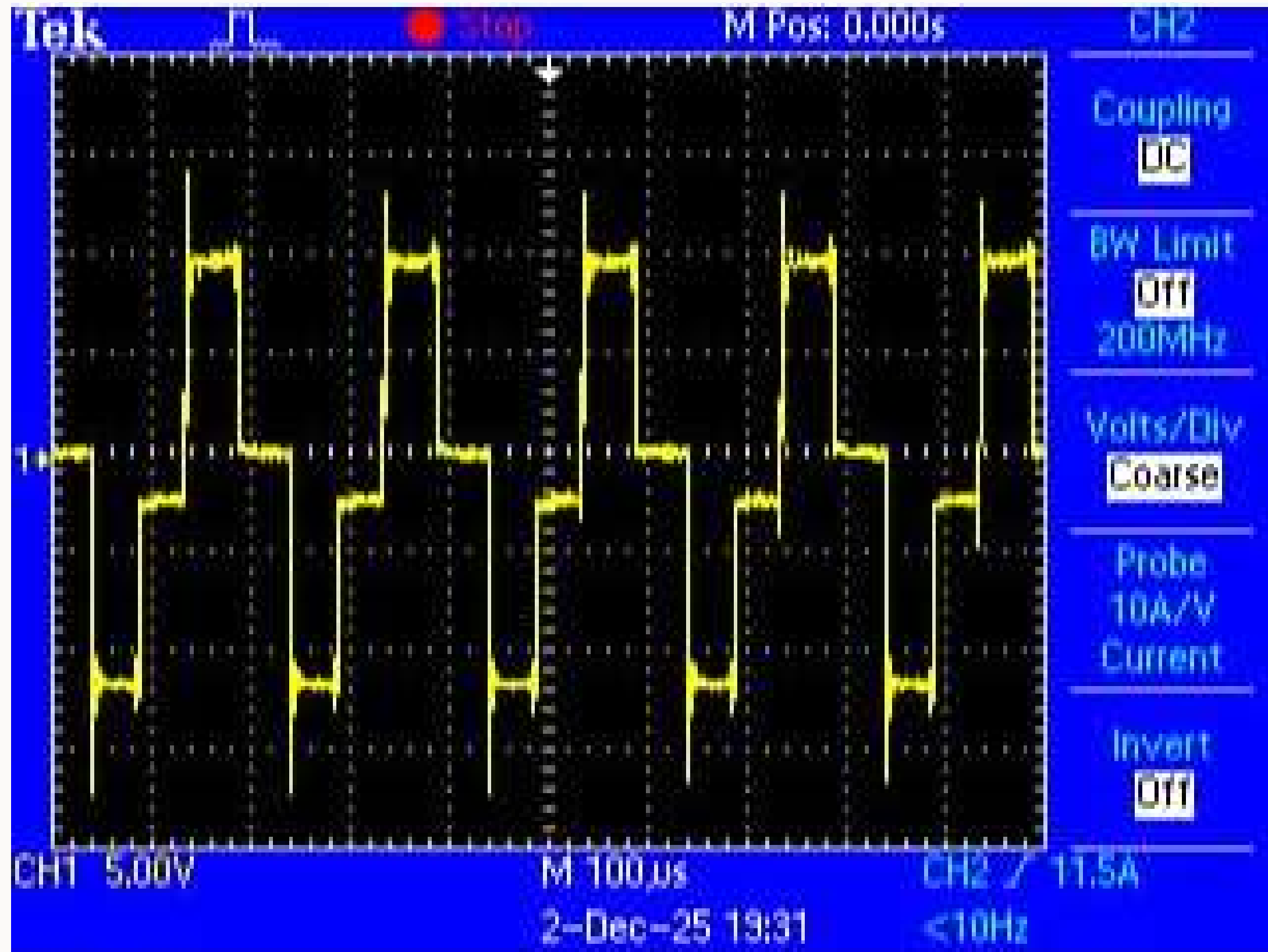


# Inverter Output waveform at 60° phase shift:





# Inverter Output waveform at 90° phase shift:



# **Applications & Impact:**

## **Primary Applications:**

- Residential EV charging stations
- Commercial charging infrastructure
- Solar-powered charging solutions
- Grid-tied smart charging systems

## **Environmental Impact:**

- Promotes renewable energy adoption
- Reduces grid dependency during peak hours
- Lowers carbon footprint of EV charging
- Supports sustainable transportation

# Challenges & Solutions:

## Technical Challenges:

- EMI Management: Solved with proper layout and filtering
- Thermal Issues: Addressed with heatsinks and airflow
- Control Complexity: Managed through modular software
- Component Selection: Optimized through simulation

## Solutions Implemented:

- Multi-layer PCB design
- Active temperature monitoring
- Hierarchical control structure
- Extensive testing and validation

# **Conclusion:**

## **Project Summary:**

- Successfully developed integrated EV charging system
- Demonstrated MPPT and PSFB control on single DSP
- Validated through hardware prototype testing

## **Key Contributions:**

- Novel DSP application in EV charging
- Cost-effective multi-source charging solution
- Framework for future smart charging systems
- Demonstration of renewable energy integration



# Acknowledgments:

We express our sincere gratitude to:

- **Prof. Rajesh Gupta** for expert guidance
- Department of Electrical Engineering for resources
- Texas Instruments for DSP support
- Family and Friends for encouragement
- All reviewers for valuable feedback

# What We Have Done:

- Designed & implemented MPPT-controlled boost converter and PSFB converter in Code Composer Studio
- Programmed Perturb & Observe MPPT algorithm and Phase-Shift control on TMS320F28379D DSP
- Developed real-time sensor feedback systems for voltage, current, and battery monitoring
- Built & tested hardware prototype integrating solar PV, power converters, and dual DSP control
- Coded embedded C with peripheral configuration (ADC, PWM, timers, interrupts) for both DSPs

# Why We Did It:

- Problem: EV charging systems lack integration between solar and efficient high-power conversion
- Challenge: Conventional systems use multiple controllers → increased cost, complexity, and reduced reliability
- Motivation: Create a unified, intelligent charging platform that maximizes solar utilization while maintaining high-efficiency charging
- Purpose: Demonstrate DSP capability for advanced power electronics control in sustainable transportation using embedded C programming

# What We Achieved:

- Final Outcome: Working dual-DSP EV charging prototype extracting max solar power and delivering efficient battery charging
- Validated Results: Oscilloscope measurements confirm stable DC bus voltage and proper phase-shift operation with dsp



*Thank You*